

ECE 6501/4501 Hardware/Software Co-Design for Machine Learning

HW1

Note: Questions 6 and 7 are required for graduate students and optional for undergraduate students.

1. Select one paper from the following four papers and answer the following questions based on the selected paper.

SqueezeNet: AlexNet-level accuracy with 50x fewer parameters and <0.5MB model size

ShuffleNet: An Extremely Efficient Convolutional Neural Network for Mobile Devices

MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Application

MobileNetV2: Inverted Residuals and Linear Bottlenecks

2. Describe the key architectural techniques used in the network, such as convolution type, bottlenecks/inverted bottlenecks, channel shuffle, residual connections, and width/depth tradeoffs.
3. Discuss the potential benefits and disadvantages of hardware implementation, such as computation, memory/data movement, parallelism, and hardware utilization.
4. Summarize the key innovations of the paper and explain why the proposed network achieves a better accuracy-efficiency tradeoff than previous approaches.
5. Provide a layer-by-layer table for the main network configuration in the paper with the following columns:

Layer # | Layer type | Input H×W | InCh | OutCh | Kernel | Stride | Output H×W

Note: Clearly state your assumptions, such as input image size and padding. Do not omit repeated or duplicated layers/blocks; list each occurrence separately. If a block contains multiple operations, list the individual operations separately.

6. Find two follow-up works that build on or improve the selected paper. For each work, briefly explain the key improvement, how it improves performance, and relevant performance metrics. Include the follow-up works as references.
7. If you were to search the architecture/hyperparameter space of the selected network:
 - a. What hyperparameters would you search? For each hyperparameter, provide potential candidate values or ranges and a search strategy.
 - b. What metrics and constraints would you optimize? Consider accuracy, latency, energy, model size, MACs/FLOPs, memory usage, and hardware area/utilization. Discuss the optimization objective and constraints.